INCH-POUND

MIL-A-82821(OS)
29 December 1989
SUPERSEDING
WS 20626A
6 August 1986
(See 6.7)

# MILITARY SPECIFICATION

## ALUMINUM, SPHERICAL (60-MICROMETERS)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

### 1. SCOPE

1.1 Scope. This specification defines one type of spherical aluminum with a nominal (target) weight-mean-diameter of 60 micrometers for use in rocket motor propellant, referred to herein as the spherical aluminum (see 6.1).

### 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6810

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

**2.1.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402-0001.)

**2.2 Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 **Material.** The spherical aluminum particles shall be dull gray in color and shall be in the form of spherical grains.

3.3 **Chemical and physical properties.** Chemical and physical properties of the spherical aluminum shall conform to table I.

TABLE I. Chemical and physical properties.

| Property | Units | Minimum | Maximum |
|---|---|---|---|
| Active aluminum | Percent | 98.0 | --- |
| Volatile content | Percent | --- | 0.10 |
| Iron content | Percent | --- | 0.40 |
| Particle size distribution | | | |
| Diameter at circumference 10 percent | micrometers | 40 | 51 |
| Diameter at circumference 50 percent | micrometers | 57 | 67 |
| Diameter at circumference 90 percent | micrometers | 70 | 83 |

3.4 **Safety**

3.4.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).



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3.4.2 **Material safety data sheet (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2 and 6.4).

3.5 **Shelf life.** The spherical aluminum shall meet the requirements of this specification for a period of 60 months from the date of quality conformance inspection (see 4.4) when packaged as specified in section 5 and stored under the storage conditions marked on the container (see 5.2.2).

3.5.1 **Shelf life extension.** The shelf life of an individual lot of spherical aluminum may be extended for an additional 60-months provided the spherical aluminum, upon retest, successfully meets the requirements of 3.3.

3.6 **Workmanship.** Workmanship shall be such that the spherical aluminum is uniform in appearance and free from visible contamination.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use (see 3.4.2).

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).



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4.3 First article inspection. First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. First article and quality conformance inspections.

| Examination and Tests | Requirement | Test Method |
|---|---|---|
| Visual examination | 3.2, 3.5 and section 5 | 4.5.1 |
| Free active aluminum test | 3.3 | 4.5.2 |
| Volatile content test | 3.3 | 4.5.3 |
| Particle size distribution test | 3.3 | 4.5.4 |
| Iron content test | 3.3 | 4.5.5 |

4.4.1 Sampling for inspection.

4.4.1.1 Lot. For inspection purposes, a lot shall consist of all the spherical aluminum subject to inspection at one time from one of the following:

a. A single batch (see 6.5.1).

b. A continuous process (see 6.5.2).

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.4.1.2 Selection of sample containers. Sample containers of the spherical aluminum shall be selected at random from each lot in accordance with table III for the inspections specified in 4.5.1 through 4.5.5. When stream sampling of material entering bulk containers is used, three to five sample increments shall be taken during loading of each container selected for sampling. The first increment shall be taken at the beginning, the last at the end and the remainder at equal intervals during loading. When stream sampling of material entering drums is used, a single sample shall be taken during loading of each drum selected for sampling.

4.4.1.3 Sample preparation. The samples selected at random from a lot shall be combined and blended to form three composite samples. An individual



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single sample shall not be split between composites. A single determination of each quality characteristic listed in table I shall be performed on each of three composite samples independently. When a lot consists of less than three containers, three samples shall be taken from each container. The sample portions thus obtained shall be grouped and blended to form three composite samples. The results of each test shall meet the requirements of table I. Failure of any composite sample to meet any requirement shall be cause for rejection of the lot.

TABLE III. Sampling for inspection.

| Number of containers in the lot | Number of containers to be sampled | Single samples per composite | | |
|---|---|---|---|---|
| | | Composite 1 | Composite 2 | Composite 3 |
| 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 |
| 3-15 | 3 | 1 | 1 | 1 |
| 16-25 | 4 | 1 | 1 | 2 |
| 26-40 | 5 | 1 | 2 | 2 |
| 41-65 | 7 | 2 | 2 | 3 |
| 66-110 | 10 | 3 | 3 | 4 |
| 111-180 | 15 | 5 | 5 | 5 |
| 181-300 | 20 | 6 | 7 | 7 |
| 301-500 | 25 | 8 | 8 | 9 |

4.4.1.4 **Sample size.** Sufficient material shall be selected at random from each sample container to perform all of the inspections specified in 4.5.1 through 4.5.5.

4.5 **Methods of inspection.** The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined in the specification. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Each sample of material shall be visually examined for conformance to the requirements of 3.2 and 3.5. Each sample shall be examined for conformance to section 5. Noncompliance with any specified requirements shall constitute cause for rejection.

4.5.2 **Free active aluminum test.** The free active aluminum tests shall be determined in accordance with the following:



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a. Weigh a 0.30 ± 0.05 gram (g) sample to the nearest 0.1 milligram (mg) and react the sample with approximately 75 milliliters (mL) of 1 Normal (N) sodium hydroxide (NaOH) solution.

b. Convert evolved hydrogen to water by passing it over cupric oxide wire, while purging with dry nitrogen, in a combustion apparatus heated to 650 to 750 degrees Celsius ($^0$C).

c. Calculate, as free active aluminum, the weight increase of a previously tared absorption tube filled with anhydrous magnesium perchlorate as follows:

$$\text{Percent free active aluminum} = \frac{(B - A)}{W} (99.85)$$

Where: A = weight of absorption tube, g

B = weight of absorption tube and moisture, g

W = weight of sample, g

99.85 = gravimetric factor for conversion of water to hydrogen to percent aluminum in this reaction

d. Record free active aluminum to nearest 0.1 percent.

4.5.3 Volatile content test. Volatile content test shall be determined in accordance with the following:

a. Weigh to the nearest 0.1 mg, 5.0 ± 0.5 g of material in a tared aluminum weighing dish.

b. Heat the dish and contents 4 hours ± 10 minutes at 105 ± 5$^0$C. Cool to room temperature in a desiccator and reweigh to the nearest 0.1 mg.

c. Calculate the weight loss of the volatiles as follows:

$$\text{Percent volatiles} = \frac{(A - B)}{W} (100)$$

Where: A = weight of weighing dish and sample, g

B = weight of weighing dish and dried sample, g

W = weight of sample, g

d. Record volatile matter to the nearest 0.01 percent.



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4.5.4 **Particle size distribution test.** The particle size distribution test, using an electronic sensing zone instrument, shall be determined in accordance with the following:

a. Use a sample size of 0.020 ± 0.004 g.

b. Use a 1 percent sodium chloride electrolyte solution (Isoton II or an approved equivalent).

c. The instrument is to be equipped with a 400 micrometer aperture and calibrated with standards having a diameter of 40 ± 8 micrometers.

d. The instrument is to be set up, calibrated and analysis run according to the procedure recommended by the manufacturer.

e. Record particle size distribution to the nearest micrometer.

4.5.5 **Iron content test.** The iron content shall be colorimetrically determined in accordance with the following:

a. Weigh to the nearest mg, a 1.0 ± 0.1 g sample and place in 500- mL volumetric flask.

b. Add 200 ± 40 mL distilled water and 10.0 mL concentrated hydrochloric acid and allow sample to completely dissolve. Dilute to volume with distilled water.

c. Pipet a 50.0-mL aliquot into a 100-mL volumetric flask and dilute to 70 ± 14 mL with distilled water.

d. Add 1 mL of 10 percent hydroxylamine hydrochloride solution and 10 mL of 0.25 percent o-phenanthroline solution and mix well.

e. Add 1 mL of 25 percent sodium citrate solution and dilute to volume with distilled water. Mix well and allow to stand at $20^{0}$C or warmer for at least 30 minutes.

f. Using a reagent blank, determine absorbance of sample solution at 510 nanometers.

g. Determine mg of iron (Fe) from a standard curve.

h. Calculate the percent iron as follows:

$$\text{Percent iron (Fe)} = \frac{A \times 50}{V \times W}$$



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Where: A = amount of iron (Fe) in aliquot obtained from standard curve, mg

V = volume of aliquot taken, mL

W = sample weight, g

50 = concentration conversion factor to percent iron

i. Record the amount of iron to the nearest 0.01 percent.

4.6 Inspection of packaging. The containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 Packing. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 Level C. The spherical aluminum shall be packaged to afford adequate protection against loss, contamination, deterioration and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.5. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 Marking.

5.2.1 Standard marking. In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 Special marking. In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

d. Container gross weight and material net weight.

e. Lot number and date of quality conformance inspection (see 3.5).

f. Shelf life and inspection/test date for extension (see 3.5). In addition, the marking shall include the following:



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"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 60 MONTH PERIOD PROVIDED THE SPHERICAL ALUMINUM, UPON RETEST, CONFORMS TO THE CHEMICAL AND PHYSICAL PROPERTY REQUIREMENTS OF MIL-A-82821."

g. Date of manufacture.

h. Date of shipment.

i. Hazardous material marking and labeling as applicable.

j. Storage conditions.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

**6.1 Intended use.** The spherical aluminum specified herein is intended for use as an ingredient in the solid propellant for the Mk 104 Standard Missile Dual Thrust Rocket Motor.

**6.2 Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article inspection is required (see 3.1 and 6.3).

d. Applicable contract provisions for Material Safety Data Sheet (see 3.4.2 and 6.4).

e. Packaging required, if different that 5.1.1.

f. Special marking, if other than as specified in 5.2.2.

g. Material safety data sheets required (see 6.4).

**6.3 First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the spherical aluminum sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include



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specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Material Safety Data Sheet (MSDS). MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed MSDSs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Definitions.

6.5.1 Batch. A batch is that amount of material made as one unit in the manufacturing process.

6.5.2 Continuous process. A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the material is accumulated for a lot.

6.6 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.7 Supersedure information. MIL-A-82821 incorporates the following engineering change proposal (ECP) and specification change notice (SCN):

| ECP | SCN |
|---|---|
| MTA003 (11/4/87) | SCN 2 (11/4/87) |

Preparing Activity:
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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-A-82821 | 2. DOCUMENT DATE (YYMMDD)<br>29 December 1989 |
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ALUMINUM, SPHERICAL 960-MICROMETERS)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
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